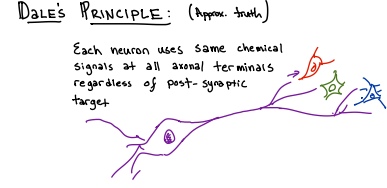
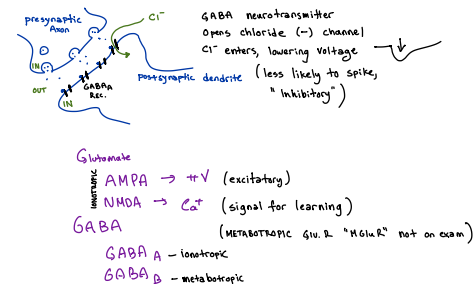


TODAY 15th Oct.
 WHY WERE WE COVERING O.D.E.s,
 EULER'S METHOD, ETC.? @
 Hippocampus, Hopfield, Delta Rule: Reprise
 ← TABLES FOR THUMB!!
 @ Exam review (continued tomorrow)

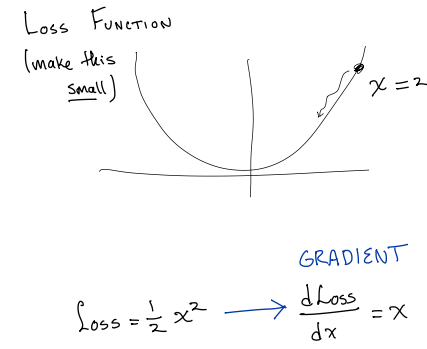


EXAMPLE SYNAPSE: IONOTROPIC GABA



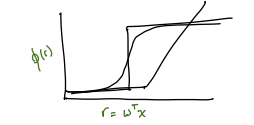
DELTA RULE

AN OPTIMIZATION PERSPECTIVE

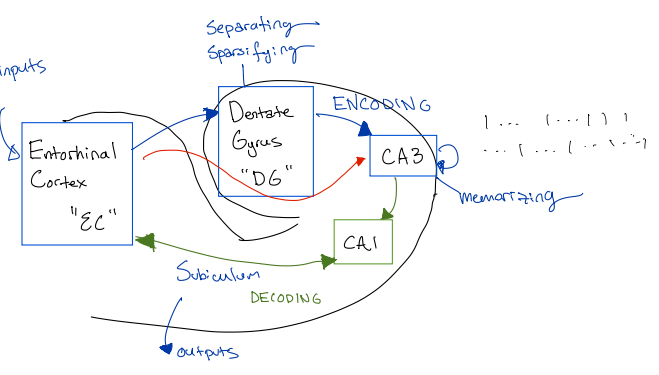
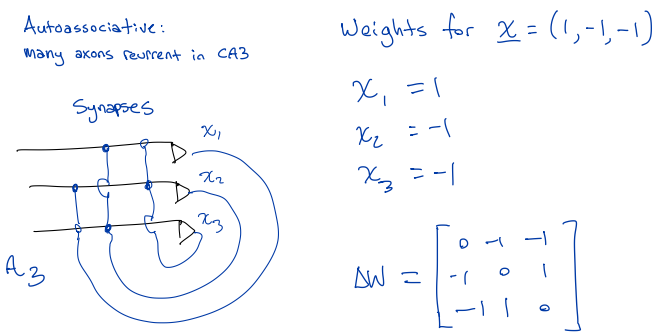


for epoch in 1...1000
 $x \leftarrow \eta (-x)$ gradient
 Learning Rate "descent"

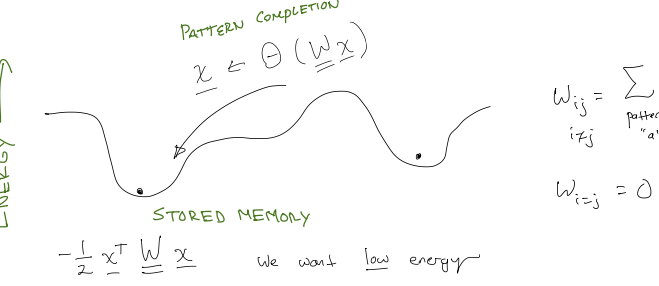
Loss $(w_1, w_2, \dots, w_n; x_1, \dots, x_n, y^*)$



HIPPOCAMPAL FORMATION & HOPFIELD



HOPFIELD-NET ENERGY $-\frac{1}{2} \underline{x}^T W \underline{x} = -\frac{1}{2} \sum_i x_i x_j w_{ij}$

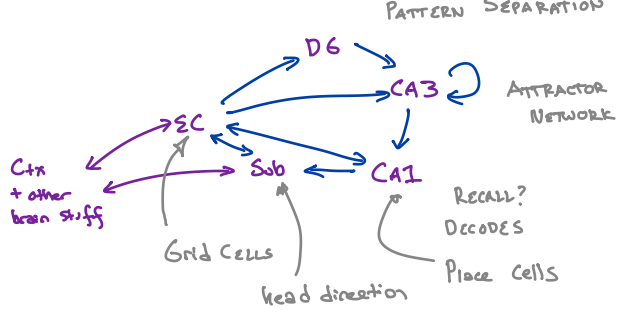


Want to remember $x_i = x_j$
 We learned $w_{ij} = +1$

Pattern energy
 $x_i \cdot x_j \cdot w_{ij}$
 $-1 \cdot -1 \cdot 1 = 1$
 $-1 \cdot -1 \cdot 1 = 1$

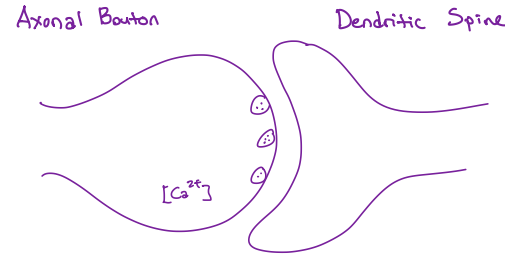
$E = -1/2$ ← better
 $E = +1/2$

Hippocampal Formation



Synaptic Depression

(Short-Term Depression STD)



let $R \in [0, 1]$ be fraction RRP available
 $1 = \text{maximal release}$ $0 = \text{nothing}$

Fraction of readily-releasable Pool Available

$\dot{R} = P_{\text{rec}} (1 - R - E) - D \cdot R \cdot \delta(t - t_d)$
 $\dot{E} = -P_{\text{rec}} E + D \cdot R \cdot \delta(t - t_d)$

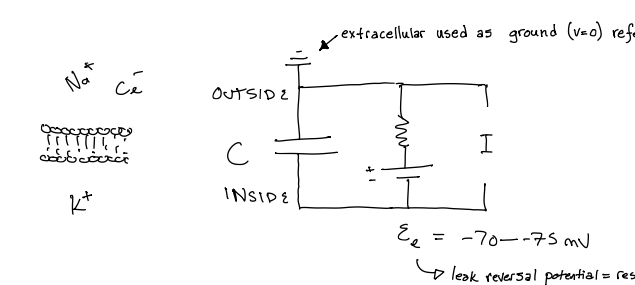
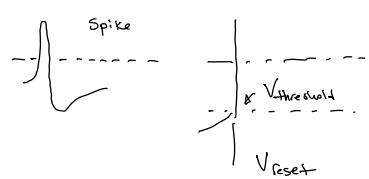
Simpler model removing the E (exhausted) state to illustrate that the Driad deltas δ are just internal jump conditions:

Contrastively
 $\tau R = 1 - R$
 When presynaptic spikes
 release $\omega \cdot R$
 $R(t^+) \leftarrow R(t^-) - D \cdot R(t^-)$

COMPUTATIONAL NEUROSCIENCE TOPICS IN CS REVIEW

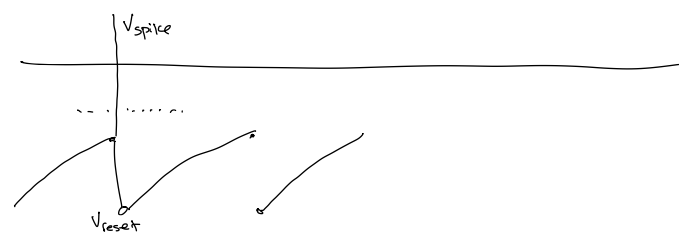
LIF Simple model of spiking + membrane voltage

key approximations in LIF:
 - Fast Time Spikes
 - $\approx$ subthreshold voltage as linear



$C \dot{V} = \frac{1}{R} (E_e - V) + I$
 $\frac{1}{R} = g$

if $V(t) > V_{\text{threshold}}$:
 $V(t) \leftarrow V_{\text{reset}}$
 (maybe same t in list of spike time)
 (updating post-synaptic conductances of target)



C: CAPACITANCE
 V: Membrane Voltage (polarized = negative)
 g: conductance = $1/\text{resistance}$ $V = IR$
 E_e or V_e : leak reversal/resting potential: where voltage goes if $I = 0$
 I: applied current

$V(t) = V_{\infty} + e^{-t/\tau} (V_0 - V_{\infty})$
 Steady state AKA V_{ss}, V
 larger = slower
 Start here
 V_0

how do we check if spike for given "I", $V_{\text{threshold}}$?

$RC \dot{V} = V_r - V + R I$
 $\tau = RC$ $V_{\infty} = V_r + R I$ $V_{\text{threshold}}$ less negative

$C \dot{V} = g (E_e - V) + I$ note $g = 1/R$ let $\tau = RC$

$\tau \dot{V} = V_{\infty} - V$ mul. by R $V_{\infty} = E_e + IR$

$V(t) = V_{\infty} + e^{-t/\tau} (V_0 - V_{\infty})$ $RC \dot{V} = E_e - V + IR$

$\tau \dot{V} = V_{\infty} - V$

$C \dot{V} = \frac{1}{R} (E_e - V) + I$
 $RC \dot{V} = E_e - V + IR$
 $\tau = RC$ $V_{\infty} = E_e$

Find spike period given constant I applied current

Assume $V(0) = V_r$ $V(t) = V_{\text{threshold}}$

$V(t) = V_{\infty} + e^{-t/\tau} (V_0 - V_{\infty})$ unknown

$V_{\infty} = V_e + R I$ source

$V_{th} = V_{\infty} + e^{-t/\tau} (V_{\text{reset}} - V_{\infty})$

$V_{th} - V_{\infty} = e^{-t/\tau} (V_{\text{reset}} - V_{\infty})$

$\ln \left(\frac{V_{th} - V_{\infty}}{V_{\text{reset}} - V_{\infty}} \right) = -\frac{t}{\tau}$

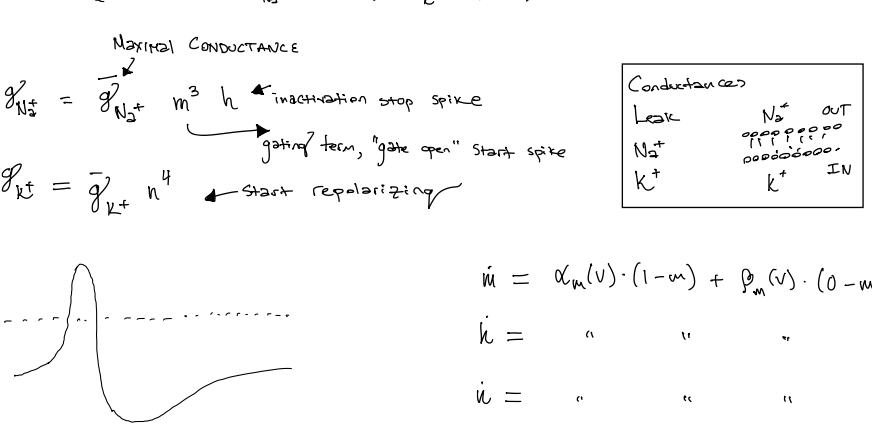
$t = -\tau \cdot \ln \left(\frac{V_{th} - V_{\infty}}{V_{\text{reset}} - V_{\infty}} \right)$ $V_{\infty} = V_r + IR$

$t = -\tau \cdot \ln \left(\frac{V_{th} - V_r - IR}{V_{\text{reset}} - V_r - IR} \right)$

4.4% f ($H_2 = \frac{1}{2}$) $\frac{1}{t}$

Hodgkin-Huxley

$C \dot{V} = g_L (E_L - V) + g_{Na}^s (E_{Na} - V) + g_K^s (E_K - V) + I$



Given this model of time dependence of K^+ gate "n"

$\tau_n(V) \cdot \dot{n} = n_{\infty}(V) - n$

Express in form $\dot{n} = \alpha \cdot (1 - n) - \beta \cdot n$
 (Take τ_n & n_{∞} as known)

Solve for α, β in terms of τ_n & n_{∞}

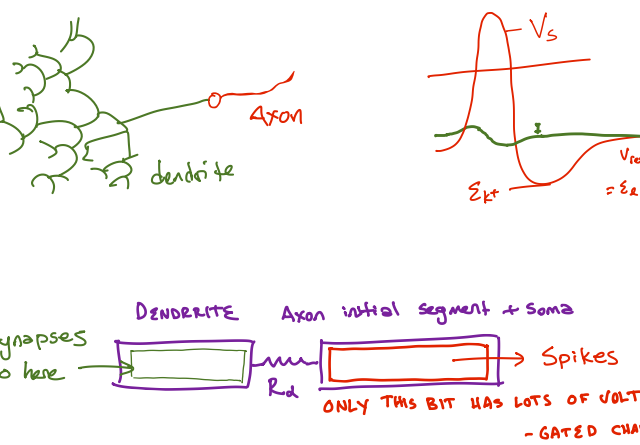
Assume $V(t)$ is constant

$\tau \dot{n} = n_{\infty} - n$
 $\dot{n} = \frac{n_{\infty}}{\tau} - \frac{1}{\tau} \cdot n$

$\dot{n} = \alpha \cdot (1 - n) - \beta \cdot n$
 $= \alpha - \alpha n - \beta n$
 $= \alpha - (\alpha + \beta) n$

$\alpha = \frac{n_{\infty}}{\tau}$
 $\alpha + \beta = \frac{1}{\tau}$
 $\beta = \frac{1}{\tau} - \alpha = \frac{1}{\tau} - \frac{n_{\infty}}{\tau} = \frac{1 - n_{\infty}}{\tau}$

Two-Compartment models



$C_d \dot{V}_d = g_d (E_d - V_d) + I_{syn} - I_{d \rightarrow s}$ ← Synaptic integration

$C_s \dot{V}_s = g_s (E_s - V_s) + I_{d \rightarrow s} + I_{spike}$ ← Spiking part

$\left(\frac{1}{R_d} (V_d - V_s) \right)$

$I_{spike} = \bar{g}_K^s n^4 (E_K - V) + \bar{g}_{Na}^s m^3 h (E_{Na} - V)$

$\tau = RC$

$\dot{V}_d = -I_{d \rightarrow s} \frac{1}{C_d} + \dots$
 $\dot{V}_s = +I_{d \rightarrow s} \frac{1}{C_s} + \dots$

